Matrix Truss Analysis with MATLAB

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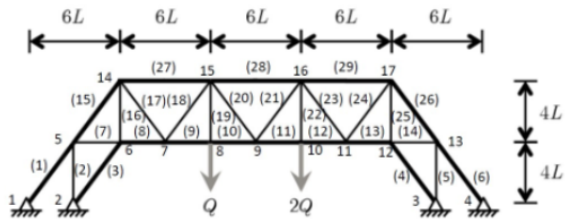

Figure 1: 3 Level Labeled Truss

0.1 Introduction

We are provided the following 29 member truss to analyze;

Our goal will be to return the following information:

- A plotted diagram showing the undeformed and deformed truss.
- The absolute extrema of the compressive and tensile stresses in all members.
- The reaction forces at the supports.

0.2 Method of Analyzing

We will approach this problem by first building our truss by setting the following parameters

Listing 1: Initialize Parameters

```
% Define nodes, elements, material, and loads
L = 1; % Length unit in feet
A = 4/144; % Bar cross-sectional area (ft\^2)
E = 30e6; % Modulus of elasticity (psi)
Q = 1000; % Applied load (lbs)
```

Matrix Setup for Truss Analysis

Next, we define the following matrices;

- **Coords** — A matrix that defines the nodes of the truss. Each row corresponds to a node with coordinates (x_i, y_i), where i is the node index. In this example, we create 17 nodes.

- **F_Applied** — A matrix where each row corresponds to a node from `Coords`. The entries (x_i, y_i) specify the direction and magnitude of the applied force at node i.
- **Bounds** — A matrix that indicates the boundary conditions (constrained or free) for each node. A value of 1 represents a fixed degree of freedom, and 0 represents a free degree of freedom.
- **U_prescribed** — A matrix specifying the prescribed displacement for each node. As an initialization, it is a matrix of zeroes of the same size as `Bounds`.
- **Connectivity** — Each row represents a truss element (member), and the two values in the row correspond to the nodes that the element connects.

These matrices provide the foundation for constructing the global stiffness matrix and performing the finite element analysis of the truss.

The coded matrices can be found in *0.4 Appendix*

0.2.1 Calculations and Visualization

Using the following idea

$$\begin{bmatrix} K_{ff} & K_{fc} \\ K_{cf} & K_{cc} \end{bmatrix} \begin{bmatrix} U_f \\ U_c \end{bmatrix} = \begin{bmatrix} P_f \\ P_c \end{bmatrix} + \begin{bmatrix} 0 \\ R_c \end{bmatrix}$$

We can partition our global matrix into several smaller matrices corresponding to free and constrained degrees of freedom that we can then use to solve for our reaction forces.

Listing 2: Assembly of Fixed and Constrained Matrices

```
U(fixedDofs) = U_presc(fixedDofs);

Kff = K(freeDofs, freeDofs);
Kfc = K(freeDofs, fixedDofs);
Ff  = F(freeDofs);

U(freeDofs) = Kff \ (Ff - Kfc * U(fixedDofs));
```

Working out our unpartitioned global matrix given above, we receive the following equations

$$\text{Free DOFs:} \quad \mathbf{K}_{ff}\,\mathbf{U}_f + \mathbf{K}_{fc}\,\mathbf{U}_c = \mathbf{P}_f \tag{1}$$

$$\text{Constrained DOFs:} \quad \mathbf{K}_{cf}\,\mathbf{U}_f + \mathbf{K}_{cc}\,\mathbf{U}_c = \mathbf{P}_c + \mathbf{R}_c \tag{2}$$

Solving these matrices will give us the following output vectors

$$\mathbf{U}^T = \begin{bmatrix} 0 & 0 & 0 & 0 & -0.0055 & -0.0062 & -0.0104 & -0.0073 & -0.0042 & 0.0009 \\ 0.0059 & 0.0021 & 0.0021 & 0.0143 & 0.0067 & -0.0102 & -0.0202 & & & \\ 0 & 0 & 0 & 0 & 0.0012 & -0.0077 & -0.0452 & -0.0756 & -0.0823 & -0.0908 \\ -0.0537 & -0.0119 & -0.0001 & -0.0140 & -0.0708 & -0.0812 & -0.0188 & & & \end{bmatrix}$$

$$\mathbf{R}_c^T = 10^3 \times \begin{bmatrix} 0.2285 & 0.3047 & 0.9847 & 1.0551 & -1.0760 & 1.4573 & -0.1372 & 0.1829 \end{bmatrix}$$

Each row of the displacement matrix $\mathbf{U}$ (Presented as $\mathbf{U^T}$ for this report) corresponds to the updated coordinates (x_i, y_i) of node i. The entries of $\mathbf{R}_c$ (Presented as $\mathbf{R_c}$ for this report) give the magnitude and direction of the reaction forces at nodes 1 through 4 in the following order: $1_x, 1_y, 2_x, 2_y, \ldots, 4_y$.

A visualization of the displacement induced by our applied loads on nodes 8 and 10, followed by a diagram of the reaction forces is shown below

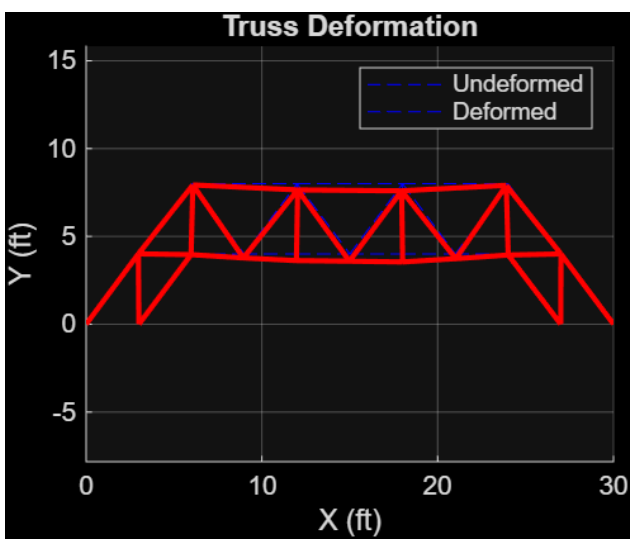

Figure 2: Plot of our Deformed vs Undeformed Truss

Finally, we will solve for the stress in each element of the truss with the following idea

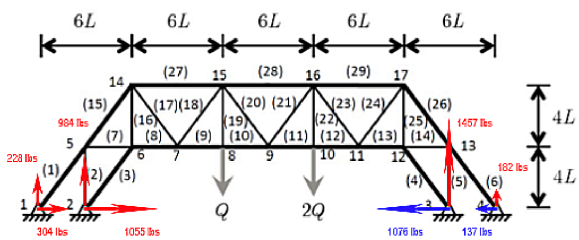

Figure 3: Truss Redrawn With Reactions

$$\sigma_e = \frac{E_e}{L_e}\begin{bmatrix} -c & -s & c & s \end{bmatrix}\mathbf{U}_e, \quad \mathbf{U}_e = \begin{bmatrix} u_{x1} \\ u_{y1} \\ u_{x2} \\ u_{y2} \end{bmatrix}, \quad e = 1, 2, \ldots, n_{\text{elem}}$$

In MATLAB, this can be done with the following code

Listing 3: Compute member stresses

```
memberStress = zeros(nElem,1);
for e = 1:nElem
    n1 = Connectivity(e,1); n2 = Connectivity(e,2);
    dx = Coords(n2,1) - Coords(n1,1);
    dy = Coords(n2,2) - Coords(n1,2);
    Le = sqrt(dx^2 + dy^2);
    c = dx/Le; s = dy/Le;
    Ae = elem_A(e); Ee = elem_E(e);
    dof = [2*n1-1 2*n1 2*n2-1 2*n2];
    Ue = U(dof);
    memberStress(e) = (Ee/Le) * [-c -s c s] * Ue;  % Axial stress =
        Force / Area
end
```

Which gives us the following output matrix, where each row corresponds to a truss member 1, 2, ..., 29

$$\text{Member Axial Stresses } \boldsymbol{\sigma}^T \text{ (10}^4 \text{ lb/in}^2\text{, + tension / - compression)} = \begin{bmatrix} -1.3712 \\ 0.9284 \\ -5.9083 \\ -6.4563 \\ -0.0812 \\ -0.8233 \\ -0.6963 \\ -4.2413 \\ 3.1015 \\ 3.1015 \\ 5.0443 \\ 5.0443 \\ -3.8129 \\ 0.0609 \\ -0.2107 \\ -4.7266 \\ 6.1190 \\ -6.1190 \\ 3.6000 \\ 1.6190 \\ -1.6190 \\ 7.2000 \\ -7.3810 \\ 7.3810 \\ -5.1650 \\ -0.9247 \\ -3.7978 \\ -8.4406 \\ -4.9834 \end{bmatrix}$$

Where the largest tensile and compressive stresses are in members 24 and 28 at 7.3810e4 lb/in^2 and 8.4406 lb/in^2

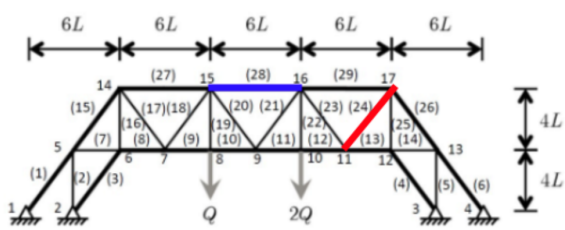

Figure 4: Enter Caption

0.3 Summary

In this project, we analyzed a truss by constructing various matrices, and translating them into a system of global matrices representing free and constrained nodal forces and displacements. We then partitioned our global system into small systems based on constrained and

free nodal parameters to then solve for displacement and reaction forces at each node, plotting both. Lastly, we used our calculations to then solve for the stress within each truss member, complete with a plot of the absolute compressive and tensile extremas

0.4 Appendix

Listing 4: Full MATLAB script

```
clc; clear; close all;

% Define nodes, elements, material, and loads
L = 1;                    % Length unit in feet
A = 4/144;                % Bar cross-sectional area (ft^2)
E = 30e6;                 % Modulus of elasticity (psi)
Q = 1000;                 % Applied load (lbs)

% Node coordinates (x, y)
Coords = [
    0 0;       %1
    3*L 0;     %2
    27*L 0;    %3
    30*L 0;    %4
    3*L 4*L;   %5
    6*L 4*L;   %6
    9*L 4*L;   %7
    12*L 4*L;  %8
    15*L 4*L;  %9
    18*L 4*L;  %10
    21*L 4*L;  %11
    24*L 4*L;  %12
    27*L 4*L;  %13
    6*L 8*L;   %14
    12*L 8*L;  %15
    18*L 8*L;  %16
    24*L 8*L;  %17
];

% Applied nodal forces (Fx, Fy)
```

```
F_Applied = [
    0 0;     %1
    0 0;     %2
    0 0;     %3
    0 0;     %4
    0 0;     %5
    0 0;     %6
    0 0;     %7
    0 -Q;    %8
    0 0;     %9
    0 -2*Q; %10
    0 0;     %11
    0 0;     %12
    0 0;     %13
    0 0;     %14
    0 0;     %15
    0 0;     %16
    0 0;     %17
];

% Boundary conditions (1=fixed, 0=free)
Bounds = [
    1 1; %1
    1 1; %2
    1 1; %3
    1 1; %4
    0 0; %5
    0 0; %6
    0 0; %7
    0 0; %8
    0 0; %9
    0 0; %10
    0 0; %11
    0 0; %12
    0 0; %13
    0 0; %14
    0 0; %15
    0 0; %16
```

```
    0 0; %17
];

% Prescribed displacements (0 for all)
U_prescribed = zeros(size(Bounds));

% Element connectivity (node pairs)
Connectivity = [
    1 5;
    2 5;
    2 6;
    12 3;
    13 3;
    13 4;
    5 6;
    6 7;
    7 8;
    8 9;
    9 10;
    10 11;
    11 12;
    12 13;
    5 14;
    14 6;
    14 7;
    7 15;
    15 8;
    15 9;
    9 16;
    16 10;
    16 11;
    11 17;
    17 12;
    17 13;
    14 15;
    15 16;
    16 17;
];
```

```
nElem = size(Connectivity,1);
elem_A = A*ones(nElem,1);
elem_E = E*ones(nElem,1);

nNodes = size(Coords,1);
GDof = 2*nNodes; % 2 DOFs per node

% Global matrices
K = zeros(GDof, GDof);                  % Global stiffness
F = reshape(F_Applied', [], 1);         % Global force vector
U_presc = reshape(U_prescribed', [], 1);

% Identify fixed and free DOFs
fixedDofs = [];
for i = 1:nNodes
    if Bounds(i,1) == 1, fixedDofs(end+1) = 2*i-1; end
    if Bounds(i,2) == 1, fixedDofs(end+1) = 2*i; end
end
freeDofs = setdiff(1:GDof, fixedDofs);

%% Assemble global stiffness matrix
for e = 1:nElem
    n1 = Connectivity(e,1); n2 = Connectivity(e,2);
    x1 = Coords(n1,1); y1 = Coords(n1,2);
    x2 = Coords(n2,1); y2 = Coords(n2,2);

    dx = x2 - x1; dy = y2 - y1;
    Le = sqrt(dx^2 + dy^2);
    c = dx/Le; s = dy/Le;

    Ae = elem_A(e); Ee = elem_E(e);

    ke = (Ae*Ee/Le) * [ c^2  c*s -c^2 -c*s;
                         c*s  s^2 -c*s -s^2;
                        -c^2 -c*s  c^2  c*s;
                        -c*s -s^2  c*s  s^2 ];

```

```
    dof = [2*n1-1 2*n1 2*n2-1 2*n2];
    K(dof,dof) = K(dof,dof) + ke;
end

% Solve for U
U = zeros(GDof,1);
U(fixedDofs) = U_presc(fixedDofs);

Kff = K(freeDofs, freeDofs);
Kfc = K(freeDofs, fixedDofs);
Ff  = F(freeDofs);

U(freeDofs) = Kff \ (Ff - Kfc * U(fixedDofs));

% Solve for RxN at 1 2 3 4
Rc = K(fixedDofs,:) * U - F(fixedDofs);

% Compute member stresses
memberStress = zeros(nElem,1);
for e = 1:nElem
    n1 = Connectivity(e,1); n2 = Connectivity(e,2);
    dx = Coords(n2,1) - Coords(n1,1);
    dy = Coords(n2,2) - Coords(n1,2);
    Le = sqrt(dx^2 + dy^2);
    c = dx/Le; s = dy/Le;
    Ae = elem_A(e); Ee = elem_E(e);
    dof = [2*n1-1 2*n1 2*n2-1 2*n2];
    Ue = U(dof);
    memberStress(e) = (Ee/Le) * [-c -s c s] * Ue;  % Axial stress =
        Force / Area
end

% Plot truss
X = Coords(:,1); Y = Coords(:,2);
scale = 5; % magnification factor
Xd = X + scale*U(1:2:end);
Yd = Y + scale*U(2:2:end);

```

```
figure; hold on; grid on;
title('Truss Deformation');
xlabel('X (ft)'); ylabel('Y (ft)');

% Plot undeformed truss
for e = 1:nElem
    n1 = Connectivity(e,1); n2 = Connectivity(e,2);
    plot([X(n1), X(n2)], [Y(n1), Y(n2)], 'b--');
end

% Plot deformed truss
for e = 1:nElem
    n1 = Connectivity(e,1); n2 = Connectivity(e,2);
    plot([Xd(n1), Xd(n2)], [Yd(n1), Yd(n2)], 'r-', 'LineWidth',2);
end

legend('Undeformed','Deformed'); axis equal;

% Results
disp('Nodal Displacements');
disp(reshape(U,2,[])');

disp('Reaction Forces at Supports');
disp(Rc);

disp('Member Axial Stresses (psi, + tension / - compression)');
disp(memberStress);
```